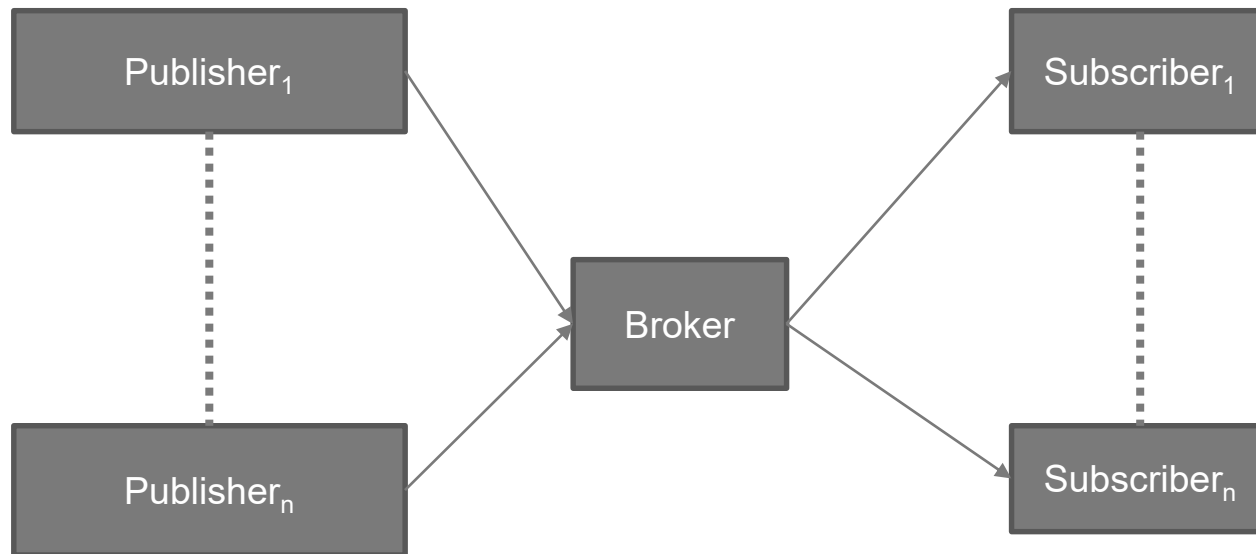


IOT MESSGING

University of Alexandria, Faculty of Engineering,
Computer and Systems Engineering Department.
CSE343 Embedded Systems Spring 2026

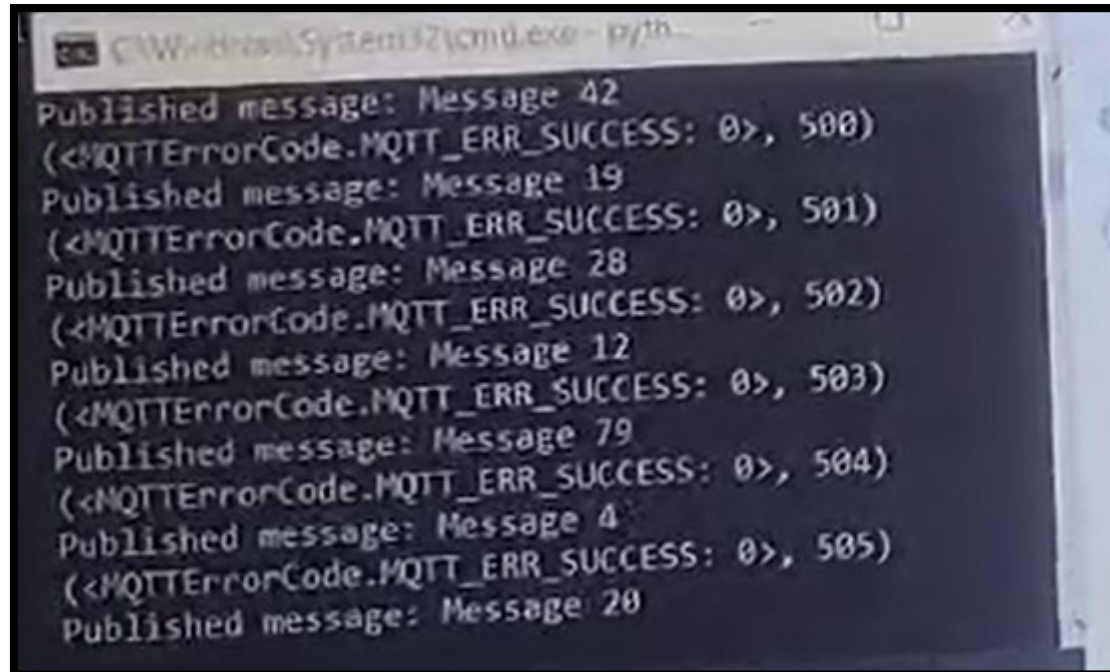
IoT Messaging Concept



Huge Data

- camera → 2 megabytes
- sensor → 20 bytes
- Camera → 100000 times sensor
- 1 second camera → 10000 seconds for imu

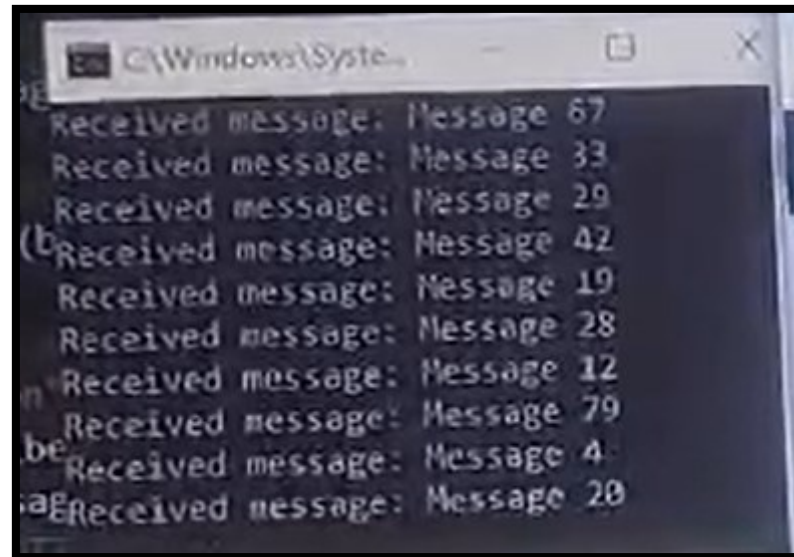
Published Message Example



A screenshot of a Windows command prompt window. The title bar reads "C:\Windows\System32\cmd.exe - python". The window contains a series of text lines representing MQTT publish operations. Each line consists of a message identifier, a success status, and a return code. The messages are: Message 42, Message 19, Message 28, Message 12, Message 79, Message 4, and Message 20. All messages show a success status of 0 and a return code of 500 through 505 respectively.

```
C:\Windows\System32\cmd.exe - python
Published message: Message 42
(<MQTTErrorCode.MQTT_ERR_SUCCESS: 0>, 500)
Published message: Message 19
(<MQTTErrorCode.MQTT_ERR_SUCCESS: 0>, 501)
Published message: Message 28
(<MQTTErrorCode.MQTT_ERR_SUCCESS: 0>, 502)
Published message: Message 12
(<MQTTErrorCode.MQTT_ERR_SUCCESS: 0>, 503)
Published message: Message 79
(<MQTTErrorCode.MQTT_ERR_SUCCESS: 0>, 504)
Published message: Message 4
(<MQTTErrorCode.MQTT_ERR_SUCCESS: 0>, 505)
Published message: Message 20
```

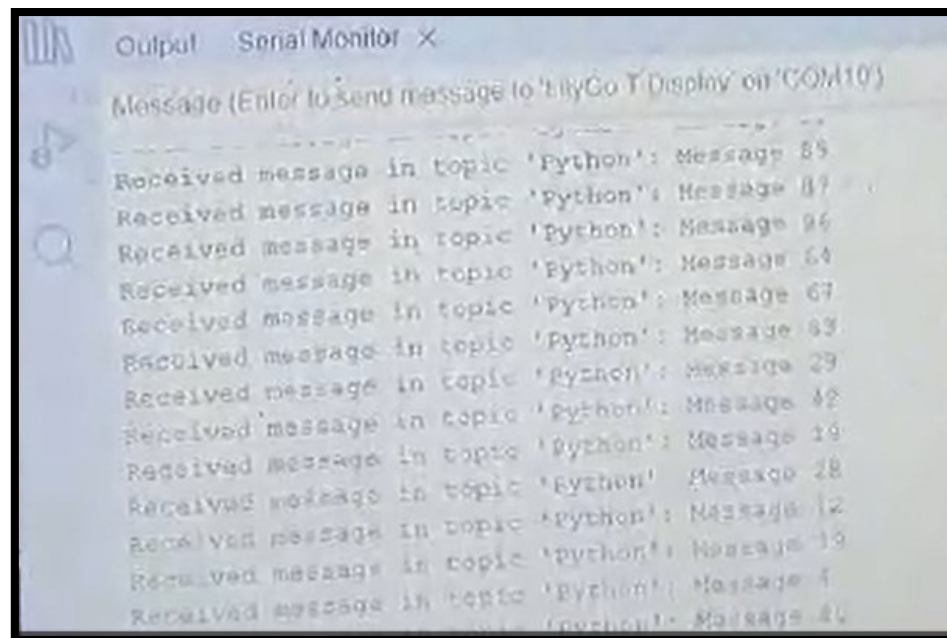
Subscribed Message Example



A screenshot of a Windows command prompt window. The title bar shows the path 'C:\Windows\System32\cmd.exe'. The window contains a list of received messages, each starting with 'Received message:'. The messages are: Message 67, Message 33, Message 29, Message 42, Message 19, Message 28, Message 12, Message 79, Message 4, and Message 20. The text is displayed in a monospaced font on a dark background.

```
Received message: Message 67
Received message: Message 33
Received message: Message 29
Received message: Message 42
Received message: Message 19
Received message: Message 28
Received message: Message 12
Received message: Message 79
Received message: Message 4
Received message: Message 20
```

Broker Example



Output Serial Monitor X

Message (Enter to send message to 'JillyGo T Display' on 'COM10')

Received message in topic 'Python': Message 89

Received message in topic 'Python': Message 87

Received message in topic 'Python': Message 96

Received message in topic 'Python': Message 64

Received message in topic 'Python': Message 67

Received message in topic 'Python': Message 63

Received message in topic 'Python': Message 29

Received message in topic 'Python': Message 42

Received message in topic 'Python': Message 19

Received message in topic 'Python': Message 28

Received message in topic 'Python': Message 12

Received message in topic 'Python': Message 13

Received message in topic 'Python': Message 1

Received message in topic 'Python': Message 40